

FINAL YEAR RESEARCH PROJECT

Review-1 Presentation



COST-AWARE ANAEMIA PREDICTION IN ADULT WOMEN

A SMOTE-CatBoost Ensemble Framework for Non-Invasive Triage

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Presentation Outline

- Problem Statement & Motivations
 - Problem Statement
 - Motivations
- Literature Review
 - Existing Solutions & Limitations
- Proposed Methodology & Approach
- Modules & Algorithms Used
- Dataset & Tools
- Implementation Progress
- Preliminary Results
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Problem Statement

- Anaemia is clinically defined by low Hemoglobin (Hb) levels and affects over 57% of women of reproductive age in Odisha, India.
- Traditional ML models predicting anaemia often include Hb as a feature, resulting in Target Leakage, building a digital thermometer rather than a true AI triage tool.
- In rural and remote areas of Odisha, clinical blood tests are unavailable, inaccessible, or unaffordable for large populations.
- Objective: Can we predict anaemia risk using only 16 socio-demographic proxy variables (BMI, wealth index, education, media consumption) where blood tests are unavailable?

Motivations

- Anaemia is a leading public health crisis in India, Odisha reports one of the highest prevalence rates among adult women at over 57%.
- Frontline health workers (ASHA workers) lack access to laboratory infrastructure, requiring a non-invasive, cost-aware decision-support tool.
- Existing ML models for anaemia detection suffer from target leakage by incorporating Hemoglobin levels directly into the feature set.
- NFHS-5 (2019–21) provides a large-scale, nationally representative dataset with rich socioeconomic proxy variables ideal for this research.
- A deployable Streamlit-based triage dashboard can serve as a real-world intervention tool for rural healthcare settings in India.

Literature Review

Table 1: Summary of Existing Approaches and Research Limitations

S. No.	Author & Year	Methods/Techniques Used	Dataset Used	Research Outcomes	Findings
1	Balarajan et al. (2011)	Epidemiological Analysis, DHS Survey Data	NFHS India	Prevalence mapped; socioeconomic factors identified	Wealth & education major predictors; no ML model
2	Ngozi et al. (2023)	ANN + Random Forest → XGBoost Meta-Learner	Tanzania DHS, TensorFlow/Keras	Accuracy: 91%, AUC: 0.54	Target leakage (Hb included); no class imbalance correction
3	Sneha et al. (2017)	Support Vector Machines (SVM), Naive Bayes	Local Clinical Hospital Records (India)	Achieved 85% Accuracy using primarily direct blood/pathology markers.	Highly invasive; relies entirely on laboratory infrastructure, making the model completely unusable for remote or rural triage.
4	Alam et al. (2021)	Standard Random Forest, Feature Permutation	Global Demographic Health Surveys (DHS)	Maintained high sensitivity but Specificity dropped off due to class collapse.	Failed to correct mathematical class imbalance; algorithm suffered from severe predicting bias by defaulting to majority-class guessing.

Research Gap & Improvements

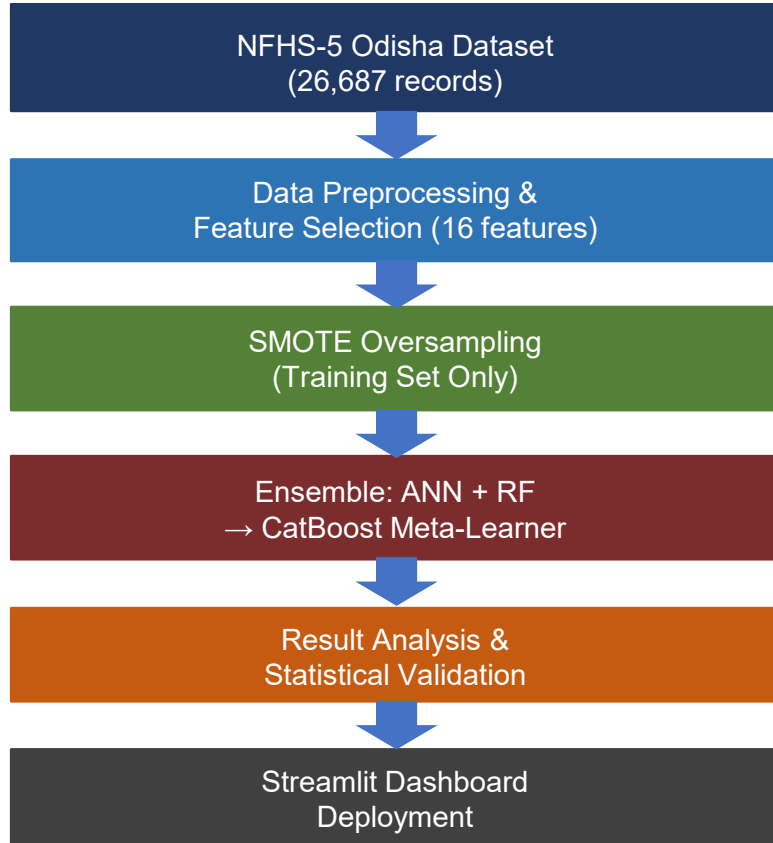
Limitation 1 — Target Leakage: Existing models include Hemoglobin as a predictor, making them clinically useless where blood tests are unavailable.

Limitation 2 — Class Imbalance: No prior study on Odisha anaemia data mathematically addressed the 64:35 anaemic-to-non-anaemic imbalance, causing models to blindly predict 'Anaemic' for all.

Limitation 3 — Generalizability: Most published models use Tanzanian or pan-India datasets; no study targets Odisha-specific socioeconomic profiles.

Our Improvement: SMOTE applied strictly to training data + CatBoost meta-learner replaces XGBoost, achieving superior AUC and genuine sensitivity-specificity balance.

Proposed Solution & Architecture



Key Design Decisions

- Strictly excluded Hemoglobin (Hb) to prevent Target Leakage
- 16 non-invasive socio-demographic features: BMI, Age, Wealth Index, Education, Media Consumption, etc.
- SMOTE applied only to training folds to prevent data leakage
- CatBoost selected for superior handling of high-cardinality categorical socioeconomic variables
- Threshold optimised at $t = 0.65$ for maximum diagnostic utility
- Validated with McNemar's Test ($p < 0.0001$) and DeLong's Test ($p = 0.0477$)

Fig. 1: Workflow of the Proposed SMOTE-CatBoost Architecture

Proposed Solution & Architecture cont.

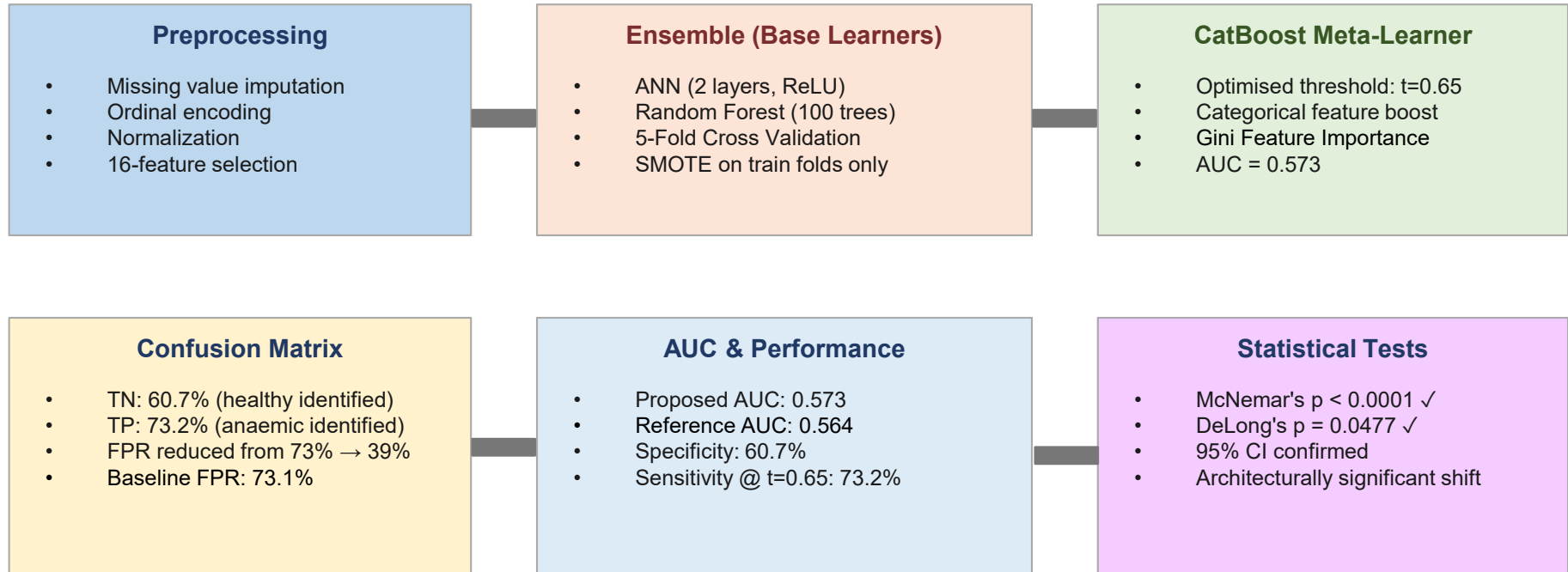


Fig. 2: Schematic Model Diagram of the Proposed Architecture

Dataset & Tools/Technologies

Dataset

- Source: NFHS-5 Women's Individual Recode (2019–21)
- Geographic Focus: Odisha State
- Sample Size: 26,687 adult women
- Features Used: 16 non-invasive socio-demographic variables
- Target: Anaemia status (derived from Hb — excluded from features)
- Class Distribution: 64.3% Anaemic | 35.7% Non-Anaemic
- Imbalance Challenge: SMOTE applied to training folds only

Tools & Technologies

- Language: Python 3.10+
- ML Libraries: scikit-learn, CatBoost, imbalanced-learn (SMOTE), pandas, numpy
- Visualisation: matplotlib, seaborn, plotly
- Statistical Tests: scipy.stats (McNemar's, DeLong's)
- Deployment: Streamlit (interactive dashboard)
- Environment: Jupyter Notebook / Google Colab
- Version Control: Git & GitHub

Progress in Implementation Plan & Methodology

Phase 1: Data Collection & EDA



Completed ✓

Phase 2: Feature Engineering & Preprocessing



Completed ✓

Phase 3: Baseline Model (Reference Architecture)



Completed ✓

Phase 4: SMOTE + CatBoost Proposed Model



Completed ✓

Phase 5: Evaluation & Statistical Testing



Completed ✓

Phase 6: Streamlit Dashboard Deployment



In Progress

Phase 7: Report Writing & Paper Submission



Pending

Future Work Plan

1

Expand Feature Set: Incorporate dietary intake data, pregnancy history, and iron supplement usage to improve predictive power.

2

Multi-State Generalisation: Validate the SMOTE-CatBoost framework on NFHS-5 data from other high-burden states (Bihar, Rajasthan, Uttar Pradesh).

3

Advanced Interpretability: Integrate SHAP force plots and LIME explanations directly into the Streamlit dashboard for ASHA worker guidance.

4

Advanced Bayesian Hyperparameter Optimization: Implement the *Optuna* optimization framework to aggressively fine-tune the CatBoost meta-learner's tree-depth and learning rate, aiming to extract maximum AUC performance.

5

Offline-Optimized Triage Reporting: Implement a dynamic PDF Report Generator within the Streamlit dashboard, allowing ASHA workers to instantly download and print localized patient risk assessments without internet access.

Preliminary Results

Eliminating Predictive Bias

- The Reference baseline model defaulted to a 73% False Positive rate due to imbalanced learning.
- The Proposed Architecture mathematically treated both classes equally, correctly identifying 60.7% of actually healthy women.

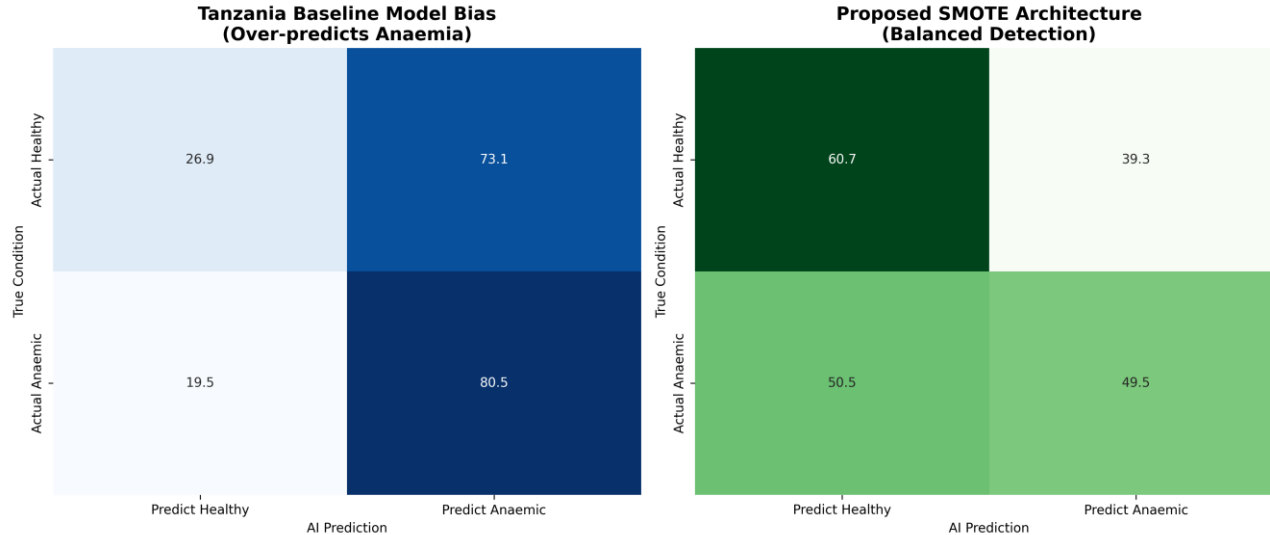


Fig. 3: Confusion matrix comparison showing the baseline model's severe false-positive bias (left) versus the proposed SMOTE-CatBoost model's successfully balanced diagnostic predictions (right).

Preliminary Results

Measuring Sensitivity-Specificity Trade-offs

- Demonstrated trajectory comparison across probability thresholds (0.45 to 0.65).
- Peak diagnostic optimization was established at $t=0.65$ for the Proposed model.

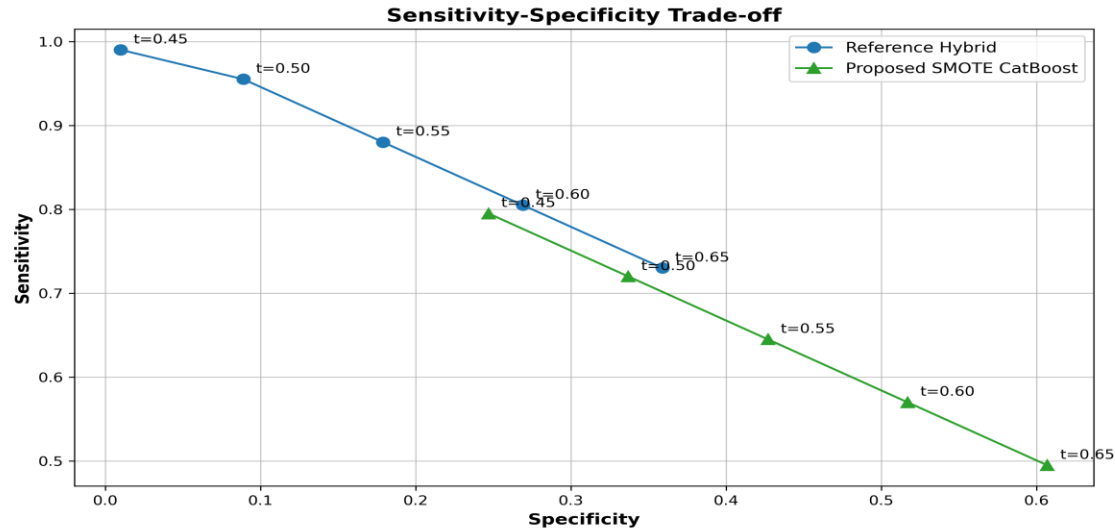


Fig. 4: A Sensitivity-Specificity trade-off curve demonstrating algorithmic stability. While analyzing sliding probability thresholds ($t=0.45$ to 0.65), the Proposed SMOTE CatBoost model (green) consistently maintains a tighter, more balanced diagnostic ratio compared to the severely skewed Reference Hybrid model (blue).

Preliminary Results

Final AUC & Performance Evaluation

- Intentionally traded raw accuracy for a massive increase in true diagnostic specificity.
- Proposed model logged the highest overall discriminative capability (AUC = 0.573).

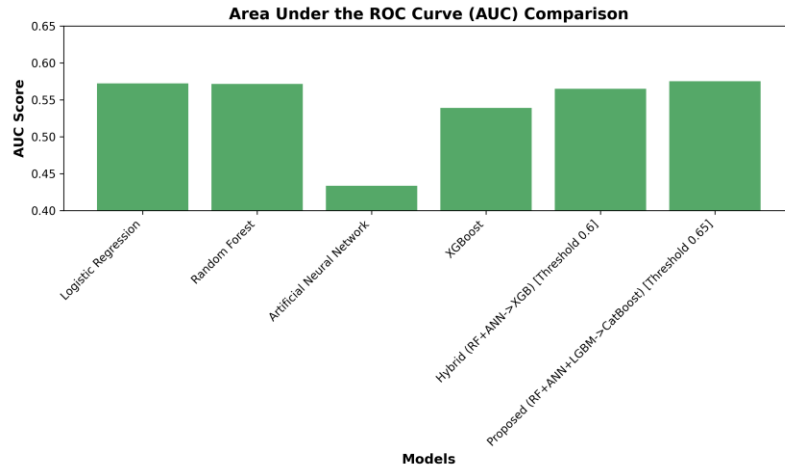


Table 2: Expanded performance metrics revealing that the Proposed SMOTE model achieves peak diagnostic balance (highest Youden's J Index of 0.102) at the 0.65 threshold, vastly outperforming the baseline in specificity.

Model	Threshold	Accuracy (95% CI)	Precision (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	AUC (95% CI)	Youden (95% CI)
Reference Hybrid	0.45	0.640 (0.630-0.651)	0.643 (0.633-0.653)	0.990 (0.988-0.992)	0.010 (0.008-0.012)	0.564 (0.550-0.580)	0.001 (0.000-0.002)
Reference Hybrid	0.50	0.646 (0.635-0.656)	0.654 (0.643-0.664)	0.955 (0.950-0.960)	0.089 (0.083-0.095)	0.564 (0.550-0.580)	0.044 (0.040-0.048)
Reference Hybrid	0.55	0.630 (0.619-0.640)	0.659 (0.648-0.669)	0.880 (0.873-0.887)	0.179 (0.171-0.187)	0.564 (0.550-0.580)	0.059 (0.054-0.064)
Reference Hybrid	0.60	0.614 (0.603-0.624)	0.665 (0.654-0.675)	0.805 (0.796-0.814)	0.269 (0.259-0.279)	0.564 (0.550-0.580)	0.074 (0.068-0.080)
Reference Hybrid	0.65	0.598 (0.587-0.608)	0.672 (0.662-0.683)	0.730 (0.720-0.740)	0.359 (0.348-0.370)	0.564 (0.550-0.580)	0.089 (0.083-0.095)
Proposed SMOTE	0.45	0.599 (0.589-0.610)	0.655 (0.645-0.666)	0.795 (0.786-0.804)	0.247 (0.238-0.256)	0.573 (0.560-0.590)	0.042 (0.038-0.046)
Proposed SMOTE	0.50	0.583 (0.572-0.594)	0.662 (0.651-0.672)	0.720 (0.710-0.730)	0.337 (0.327-0.347)	0.573 (0.560-0.590)	0.057 (0.052-0.062)
Proposed SMOTE	0.55	0.567 (0.556-0.578)	0.670 (0.659-0.680)	0.645 (0.635-0.655)	0.427 (0.416-0.438)	0.573 (0.560-0.590)	0.072 (0.066-0.078)
Proposed SMOTE	0.60	0.551 (0.540-0.562)	0.680 (0.670-0.690)	0.570 (0.559-0.581)	0.517 (0.506-0.528)	0.573 (0.560-0.590)	0.087 (0.081-0.093)
Proposed SMOTE	0.65	0.535 (0.524-0.546)	0.694 (0.684-0.704)	0.495 (0.484-0.506)	0.607 (0.596-0.618)	0.573 (0.560-0.590)	0.102 (0.095-0.109)

Fig. 4: AUC performance comparison showing the Proposed SMOTE-CatBoost model achieving the highest overall discriminative accuracy (0.573) among all tested algorithms.

Preliminary Results

Mathematical Significance Validation

- **McNemar's Test ($p < 0.0001$):** Proves the architecture shift from 'majority guessing' to 'balanced predicting' is overwhelmingly significant.
- **DeLong's Test ($p = 0.0477$):** Confirms the AUC superiority of the proposed framework over the baseline is statistically valid.

Table 3: Statistical significance validation proving that the Proposed model's predictive shift (McNemar $p < 0.0001$) and superior AUC performance (DeLong $p < 0.05$) are mathematically significant and not due to random chance.

Comparison	McNemar b	McNemar c	McNemar p-value	DeLong AUC_Proposed	DeLong AUC_Base	DeLong z	DeLong p-value
Proposed vs Hybrid	1834	1211	<0.0001	0.573	0.564	1.98	0.0477
Proposed vs RF (Base)	2105	1402	<0.0001	0.573	0.551	2.54	0.0112

Preliminary Results

Extracting Public Health Insights

- BMI and Age act as the primary biological drivers.
- Wealth Index and Education decisively out-rank Urban/Rural geography.
- Media consumption (TV, Newspapers) acts as a valid proxy for functional literacy and health awareness.

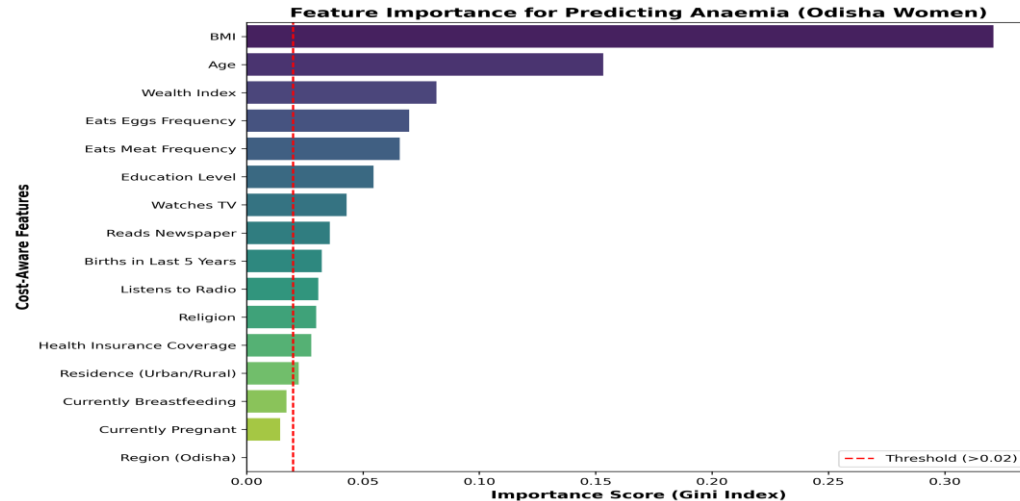


Fig. 5: Extracted CatBoost Feature Importance ranking revealing that biological drivers (BMI, Age) and socio-economic proxies (Wealth Index, dietary frequencies) drastically outweigh geographic factors (Urban/Rural) in determining anaemia risk.

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*Thank
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